2 Dry thermodynamics: atmospheric statics, first law, specific heats

Reading for Day 2: Pierrehumbert Principles of Planetary Climate (PPC), Sections 2.1-2.3, 2.5.

Just a reminder that I am *not* following this textbook closely (or in the same order), but I think that it is the best possible reference for the material we will be covering. I think it is especially useful to have multiple viewpoints on the same topic – mine is more that of a planetary scientist[6], while Prof. Pierrehumbert has a deeper view encompassing the legacy of Earth and planetary climate alongside an understanding of the physical principles.

2.1 Thermodynamic description of air

2.1.1 Knudsen number

For most atmospheric phenomena of interest in this course, we are focused on scenarios where the relevant length scales are much larger than the typical mean free path of atmospheric species. This can be quantified with the Knudsen number,

$$\mathrm{Kn} = \frac{\lambda}{\mathcal{L}}, \tag{2.1}$$

where $\lambda = 1/(n\sigma)$ is the mean free path of the gas with n the number density and σ the elastic scattering (collision) cross section, and $\mathcal{L}$ is a typical lengthscale of interest. In our classroom, $n \sim 10^{25}\mathrm{m}^{-3}$ and typically $\sigma \sim 10^{-19}\mathrm{m}^2$ – as a result $\lambda \sim 10^{-6}$ m $= 1\mu$m. Compare this to any lengthscale we might be interested in considering atmospheric flows – whether the small-scale motions of clouds, or large-scale atmospheric dynamics including jets and waves (that you'll learn from Prof. Woolings), and you will find that $\mathrm{Kn} \ll 1$. This implies that we can describe the atmosphere for our purposes with the continuum equations of mechanics and thermodynamics, and I will do so throughout these notes.

2.1.2 Equation of state

An equation of state is a relationship between thermodynamic variables, typically including pressure, temperature, and density. The ideal gas law generally suffices as an equation of state for atmospheres[7], which relates pressure p, temperature T, and number density (in moles per volume) as

$$p = nR_uT, \tag{2.2}$$

where $R_u = 8.314$ J mol^{-1} K^{-1} is the universal gas constant. Often, atmospheric scientists define a specific gas constant $R = R_u/\mu$, where μ is the mean molecular weight (mass per mole) of the atmospheric mixture of interest. Using the specific gas constant, we can write the ideal gas law as

$$p = \rho RT, \tag{2.3}$$

[6]Planetary scientists typically are trained to have a very diverse skillset encompassing planetary physics, chemistry, and geology. I am knowledgeable and able to hold discussions on many topics, but an actual expert in very few.

[7]Note that it does not suffice at sufficiently high pressures, but for the purposes of these notes we will differentiate the "atmosphere" from the "interior" through the use of standard approximations that can be used to study Earth's atmosphere.

Atmosphere	μ (kg/mole)	**R** (J/kg K)
Earth	0.029	287
Jupiter	0.0023	3600
CO_2	0.044	189

Table 1: Mean molecular weights and specific gas constants for common planetary atmospheres.

where here ρ is the mass density (mass per volume) of the air. An equivalent form you may see in atmospheric contexts uses the specific volume $v = 1/\rho$,

$$pv = RT. \tag{2.4}$$

Table 1 shows the mean molecular weight and specific gas constant for some specific atmospheric cases of interest.

Note that individual components of air each obey their own ideal-gas type relationship

$$p_i = \rho_i R_i T, \tag{2.5}$$

where p_i is the partial pressure, which is the pressure that component i would have in isolation at the same density and temperature as the mixture, and ρ_i and R_i are similarly the density and specific gas constant of component i (and one can also define the number density of species i as $n_i = p_i/RT$). The total pressure is then simply the sum of all partial pressures,

$$p = \sum_i p_i. \tag{2.6}$$

2.1.3 Expressing concentrations

Because air is compressible, the absolute concentrations (i.e., densities, ρ_i or n_i) can change even if the number of moles of a given species within an air parcel is fixed[8]. As a result, we typically express the concentration of a species i relative to the total concentration.

The simplest way to express the relative concentration of a species is the mole fraction,

$$f_i \equiv \frac{n_i}{n}. \tag{2.7}$$

However, the mole fraction is not a useful quantity when taking into account condensation or other processes that change the relative abundances of species in air (e.g., when water vapour rains out on Earth, then the mole fraction of N_2, O_2, CH_4, CO_2, and all other trace species also changes). We can deal with this by using dry air as our reference point, rather than to the entire mixture. This is the mixing ratio, and we can express this either in mass

$$q_i = \frac{\rho_i}{\rho_d} = \frac{\mu_i}{\mu_d}\frac{n_i}{n_d} = \frac{\mu_i}{\mu_d}\frac{p_i}{p_d}, \text{ mass mixing ratio,} \tag{2.8}$$

where for the last equivalence we used the ideal gas law, or in volume (equivalently number)

$$\xi_i = \frac{n_i}{n_d}, \text{ volume mixing ratio,} \tag{2.9}$$

where in the above subscript d refers to the dry component of the mixture.

[8]Note that the number of moles could of course also change due to condensation or chemical reactions.

2.2 Hydrostatic balance

2.2.1 Derivation

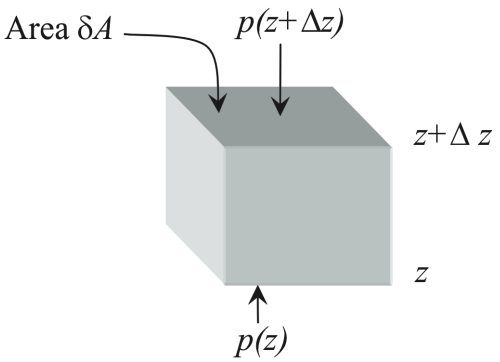


Figure 2.1: Schematic of a parcel of gas in hydrostatic equilibrium.

Figure 2.1 shows a parcel of gas held between z and $z+\Delta z$ at pressure $p(z)$ at the bottom and $p(z+\Delta z)$ at the top. Newton's second law ($F = ma$) implies that the change in pressure across Δz is the weight per area of the gas:

$$[p(z+\Delta z) - p(z)]\delta A = -\rho g \Delta z \delta A \ , \tag{2.10}$$

where $g = GM_p/(R_p+z)^2 \approx GM_p/R_p^2$ is the gravity, typically assumed to be constant in the thin-atmosphere approximation[9], and G is the universal gravitational constant. Thus,

$$\frac{p(z+\Delta z) - p(z)}{\Delta z} = -\rho g \ , \tag{2.11}$$

and given that

$$\lim_{\Delta z \to 0} \frac{p(z+\Delta z) - p(z)}{\Delta z} = \frac{dp}{dz} \ , \tag{2.12}$$

we can write the expression for hydrostatic balance

$$\frac{dp}{dz} = -\rho g \ . \tag{2.13}$$

Importantly, note that Equation (2.13) holds well to very good approximation *locally* on Earth as well as globally – this is important not just for its vertical structure but also geophysical fluid dynamics, as will be covered in Prof. Woolings' section of this course.

2.2.2 Implications of hydrostatic balance

Note that if we integrate hydrostatic balance (Equation 2.13) vertically assuming constant gravity, we find that the surface density (i.e., mass per area) in a given atmosphere column increases with pressure and decreases with gravity:

$$\frac{\text{mass}}{\text{area}} = \Sigma = \frac{p}{g}. \tag{2.14}$$

[9]Note that this approximation breaks down for planets with thick envelopes, but is suitable for most models of Earth.

We can further substitute the ideal gas law (Equation 2.3) into the expression for hydrostatic equilibrium and integrate to find

$$\begin{aligned} \frac{1}{p}\frac{dp}{dz} &= -\frac{g}{RT} \\ \ln\left(\frac{p}{p_0}\right) &= -\int_{z_0}^{z}\frac{g}{RT}dz' \\ p = p_0\ \exp\left(-\int_{z_0}^{z}\frac{g}{RT}dz'\right) &= p_0\ \exp\left(-\int_{z_0}^{z}\frac{dz'}{H}\right), \end{aligned} \tag{2.15}$$

where $H = RT/g$ is the pressure scale height. For an isothermal atmosphere, the expression for pressure on height from hydrostatic equilibrium simplifies to (where here p_0 is the surface pressure where $z = 0$)

$$p = p_0 e^{-z/H}, \tag{2.16}$$

and thus the pressure (and density, for an isothermal atmosphere) decreases with increasing height over a characteristic e-folding distance of H. As a result, H is often termed the "isothermal" scale height, as it indeed serves as a characteristic e-folding distance in the vertical in the isothermal limit[10]. Note that $H \approx 8.5$ km for Earth.

2.2.3 Vertical structure of planetary atmospheres

Figure 2.2 shows that as expected from our integration of hydrostatic balance above, in Earth's atmosphere the dependence of pressure and density on height is approximately exponential. The slight deviations from a pure exponential decrease are due to changes in the temperature with height, demonstrating that an isothermal approximation is insufficient to calculate the detailed pressure or density profile of a typical planetary atmosphere.

However, the dependence of temperature on height is not so straightforward – for Earth there are multiple atmospheric levels, defined by the temperature gradient. The region near the surface where the temperature decreases with increasing height is known as the *troposphere*, with the region overlying that where the temperature increases with height known as the *stratosphere*[11]. Near-surface tropospheres and overlying stratospheres are ubiquitous in our Solar System, as shown in Figure 2.3. We'll next dive into the fundamental thermodynamics that controls these vertical temperature profiles.

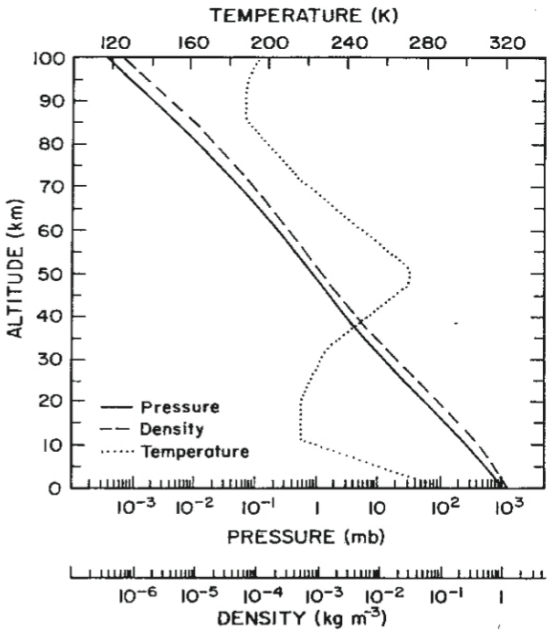
2.3 First law of thermodynamics

The first law of thermodynamics is a statement of energy conservation – namely, that the rate of change in the internal energy of a system is the sum of the working and heating rates

$$\frac{dU}{dt} = Q + W, \tag{2.17}$$

[10] If one is considering a localized enough region, then the isothermal limit applies – hence why the local isothermal scale height can be a useful quantity.

[11] One can further demarcate regions above this as mesosphere, thermosphere, exosphere, etc., but in C5 we largely focus on the troposphere and stratosphere given that these are the regions most relevant to weather prediction and present-day climate.



Global-mean pressure (solid), density (dashed), and temperature (dotted), as functions of altitude.
Source: U.S. Standard Atmosphere (1976).

Figure 2.2: U.S. standard atmosphere of Earth. Shown are pressure, density, and temperature profiles for typical Earth climate conditions. Note that the pressure and density x-axis is logarithmic.

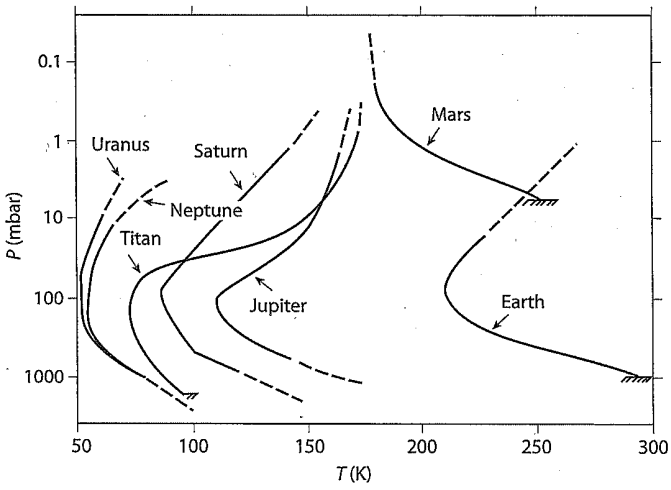


Figure 2.3: Atmospheric temperature-pressure profiles of planets in our Solar System with significant atmospheres (not including Venus), along with Titan. Figure adapted from Seager (2010).

where $U(V,T)$[12] is the internal energy, Q is the heating rate (which represents interactions with environments at different temperature through processes such as conduction and radiation), and W is the working rate (which represents a mechanical exchange of energy with the environment). For an atmosphere, the working rate is dominated by the expansion work done *on* the system

$$W = -p\frac{dV}{dt}, \tag{2.18}$$

[12]Here neglecting chemical potential.

where V is the volume of the system. Thus, for an atmosphere we can write the first law of thermodynamics as

$$\frac{dU}{dt} = Q - p\frac{dV}{dt}. \tag{2.19}$$

For studies of atmospheres, an alternate version of the first law of thermodynamics is often used with enthalpy $H = U + pV$ instead of internal energy, implying $U = H - pV$ and thus $\frac{dU}{dt} = \frac{dH}{dt} - p\frac{dV}{dt} - V\frac{dp}{dt}$. Substituting in, we can re-write the first law as

$$\frac{dH}{dt} = Q + V\frac{dp}{dt}. \tag{2.20}$$

Note that a constant p, constant T process has constant H but does *not* have constant U because work can be done on the gas.

2.4 Specific heats

The chain rule can be used to re-state the change in internal energy as

$$\frac{dU}{dt} = \frac{\partial U}{\partial T}\frac{dT}{dt} + \frac{\partial U}{\partial V}\frac{dV}{dt}. \tag{2.21}$$

We can define the heat capacity at constant volume C_v to be

$$C_v \equiv \left(\frac{\partial U}{\partial T}\right)_v, \tag{2.22}$$

which is a *specific heat* (c_p) when defined per mass or per mole. Note that for an ideal gas changes in internal energy due to changes in volume can be neglected[13], so we can write the change in internal energy as

$$\frac{dU}{dt} = C_v\frac{dT}{dt}. \tag{2.23}$$

We can equivalently perform the chain rule on the change in enthalpy with time to express

$$\frac{dH}{dt} = \frac{\partial H}{\partial T}\frac{dT}{dt} + \frac{\partial H}{\partial p}\frac{dp}{dt}. \tag{2.24}$$

We can similarly define the heat capacity at constant pressure C_p to be

$$C_p \equiv \left(\frac{\partial H}{\partial T}\right)_p. \tag{2.25}$$

For an ideal gas, the enthalpy changes due to pressure changes are negligible (i.e., , the second term on the right hand side of Equation (2.24) is zero), so we can write the change of enthalpy as

$$\frac{dH}{dt} = C_p\frac{dT}{dt}. \tag{2.26}$$

[13]This is because the changes in internal energy caused by changes in volume would arise if inter-molecular interactions were important. In an ideal gas, inter-molecular interactions are by definition not important.

We can then relate the heat capacities at constant pressure and volume as

$$C_p = C_v + \frac{\partial U}{\partial V}\frac{\partial V}{\partial T} + p\frac{\partial V}{\partial T}. \tag{2.27}$$

For an ideal gas, $\frac{\partial U}{\partial V} \equiv 0$. Thus, for an ideal gas the heat capacity and specific heat are related as

$$\begin{aligned} C_p &= C_v + NR_u, \\ c_p &= c_v + R_u, \end{aligned} \tag{2.28}$$

where N is the number of moles in the system and the bottom equation is written per molecule. If we instead divide through to write this relationship in per mass units, we find

$$c_p = c_v + R, \tag{2.29}$$

where here R is the *specific* gas constant, not the universal gas constant.

The heat capacity C_v per molecule is $k_B/2$ per degree of freedom, and note that the universal gas constant is related to the Boltzmann constant[14] as $R_u = N_A k_B$ with $N_A = 6.022 \times 10^{23}$ mol^{-1} Avogadro's number. Thus, the specific heat capacity c_v can also be expressed as $R_u/(2)$ per degree of freedom. For diatomic molecules, there are 3 translational and 2 rotational degrees of freedom, leading to $c_v = 5/2R_u$ and $c_p = 7/2R_u$.

Questions for the class:

1. Why isn't there a third rotational degree of freedom for a diatomic molecule?

2. And why have we neglected the vibrational degrees of freedom here?

Two common combinations of specific heats are

$$\begin{aligned} \gamma &\equiv \frac{c_p}{c_v}, \\ \kappa &\equiv \frac{R}{c_p}, \end{aligned} \tag{2.30}$$

where for Earth air $\gamma \approx 1.4$ and $\kappa = 2/7 \approx 0.286$. The specific heat capacities and specific heat ratios of relevant atmospheric gases are shown in Figure 2.4.

[14]Prof. Pierrehumbert's textbook even says "the Boltzmann constant should not be considered a fundamental constant of the Universe. It is just a unit conversion factor."

	H_2O	CH_4	CO_2	N_2	O_2	H_2	He	NH_3
Crit. point T	647.1	190.44	304.2	126.2	154.54	33.2	5.1	405.5
Crit. point p	221.e5	45.96e5	73.825e5	34.0e5	50.43e5	12.98e5	2.28e5	112.8
Triple point T	273.15	90.67	216.54	63.14	54.3	13.95	2.17	195.4
Triple point p	611.	.117e5	5.185e5	.1253e5	.0015e5	.072e5	.0507e5	.061e5
L vap(b.p.)	22.55e5	5.1e5	–	1.98e5	2.13e5	4.54e5	.203e5	13.71e5
L vap(t.p.)	24.93e5	5.36e5	3.97e5	2.18e5	2.42e5	??	??	16.58e5
L fusion	3.34e5	.5868e5	1.96e5	.2573e5	.139e5	.582e5	??	3.314e5
L sublimation	28.4e5	5.95e5	5.93e5	2.437e5	2.56e5	??	??	19.89e5
ρ liq(b.p.)	958.4	450.2	1032.	808.6	1141.	70.97	124.96	682.
ρ liq(t.p.)	999.87	??	1110.	??	1307.	??	??	734.2
ρ solid	917.	509.3	1562.	1026.	1351.	88.	200.	822.6
$c_p(0C/1bar)$	1847.	2195.	820.	1037.	916.	14230.	5196.	2060.
$\gamma(c_p/c_v)$	1.331	1.305	1.294	1.403	1.393	1.384	1.664	1.309

Table 2.1: Thermodynamic properties of selected gases. Latent heats of vaporization are given at both the boiling point (the point where saturation vapor pressure reaches $1bar$) and the triple point. Liquid densities are given at the boiling point and the triple point. For CO_2 the 'boiling point' is undefined, so the liquid density is given at $253K/20bar$ instead. Note that the maximum density of liquid water is $1000.00kg/m^3$ and occurs at $-4C$. Densities of solids are given at or near the triple point. All units are mks, so pressures are quoted as Pa with the appropriate exponent. Thus, $1bar$ is written as $1e5$ in the table.

Figure 2.4: Relevant thermodynamic data for common atmospheric gases. Table adapted from an early version of Pierrehumbert (2010) (see his Table 2.1).